Comparing Quantities

07



Whenever we think of comparing quantities of the same type we first think of ratio and proportion. What is a ratio? Is proportion different from ratio? Let us look at some examples to understand this.

Children of a class were measuring the distance from their ankle to the knee using their handspan.

Neelam : the distance of my knee from my ankle is 2 hand spans and 4 fingers

Kiran : this distance for me was also 2 hand spans and 4 fingers

Then each child one by one said that the distance for me is also 2 hand spans and 4 fingers.

Measure the distance from your knee.

Can we say that ratio of the length of the hand span to the distance from the knee to the ankle remains the same for all?

One More Measurement

Mahi said let's measure the length of the arm (shoulder to the wrist) using hand spans.



Saurabh : This length is also 2 hand spans and 4 fingers

Mahi : It is the same for me

The measure of the distance from the shoulder to the wrist was 2 hand spans and 4 fingers for all children.

You also measure this distance. We can say that the ratio of the length of the hand span to the distance from the shoulder to the wrist remains the same for all.

We therefore see that the ratio of the length of the hand span to the distance from the shoulder to the wrist and its ratio to the distance from the knee to the ankle is the same for all. We say that these have the same proportion.

Look at these also

If there are three black and nine white balls in a bag then we can say;

1. The number of black balls is one third $\left(\frac{1}{3}\right)$ the number of white balls

2. White balls are three times the number of black balls

Black balls	**:**	**White balls**
3	:	9
1	:	3

Similarly if the height of a banana plant is 20 cm and that of a mango plant is 60 cm then the ratio of the heights is-

Height of the banana plant	**:**	**Height of the mango plant**
20	:	60
1	:	3



This is the same ratio as we saw between black and white balls. These are therefore proportional.

Think and Discuss

Give five examples from your daily experience where you find definite ratio or proportion between different quantities.

Rehana studies in class 8 and Farida is in the class 9. Both of them got their results for the maths exams. They went and told their results to their mother.

Rehana : I got 80 marks out of 100

Mother : Ok! And Farida you?

Farida : I got 110 marks out of 150.

Rehana : Wow! Didi has better result compared to me.

Mother : How?

Rehana : Didi has got more marks than me, therefore her result is better.

Farida : Oh yes! I've got 110 marks and Farida has got only 80



Mother : But Farida, Rehana has got 80 marks out of hundred and you have got 110 out of 150, so how do you know your result is better than Rehana's?

Farida : I did the following-

Rehana got- 80 marks

And I got- 110 marks

and then I subtracted 80 marks from 110.

$110 - 80 = 30$

So I got 30 more marks than Rehana, hence, my result is better.

Mother : But the total marks for your cases are different, therefore we cannot say who has a better result. If the look at the ratio of the marks obtained against total marks, then we can say-

Rehana got- $\frac{80}{100} = \frac{8}{10} = \frac{4}{5}$ and

Farida got- $\frac{110}{150} = \frac{11}{15}$



To compare the two ratios we make the denominators the same, we get then

Rehana's marks- $\frac{4}{5} \times \frac{3}{3} = \frac{12}{15}$ that is 12 out of 15

Farida's marks- $\frac{11}{15}$ that is 11 out of 15

And because $\frac{12}{15}$ is more than $\frac{11}{15}$, Rehana's result is better.

Farida : Can we do this some other way?

Mother : Yes! We can also use percentages for this.

What are Percentages?

Percentage is another way of comparing quantities. When we compare on the basis of percentages (%) we assumed the total to be 100.

For example if out of 12 balls in a bag 3 are black then the percentage of black balls is $\frac{3}{12} \times 100 = 25\%$

The percentage of white balls is $\frac{9}{12} \times 100 = 75\%$

We can then say that 25% of the balls are black and 75% are white.

Percentage means per 100 and is shown by % sign.

Think and Discuss

Consider the following examples carefully and discuss which way of comparison, ratio or percentages would be better?

1. The mixture for Idli contains

$\frac{1}{3}$ urad lentil $\frac{2}{3}$ rice	33.3% urad lentil 66.6% rice



2. The distribution of children in the class is

Girls : Boys 3 : 2	60% Girls 40% Boys

Application of Percentage

Discount

Deepak wants to buy some material for the school.

He went to the market and saw a sign board outside the stationery shop saying discount of 6% on all you buy.



Deepak bought a packet of pencils and asked the shopkeeper the price?

Shopkeeper : The price is Rs.50 but due to 6% discount you pay 47.

Deepak : Ok! How did you calculate that?

Shopkeeper : The discount is calculated on the marked price of the item, therefore when we sell the item we deduct the discount amount and therefore the selling price is less than the marked price.

Discount = Marked Price – Selling price

The discount is expressed in percentage and calculated on the marked price.

EXAMPLE-1. The price of a shirt is Rs 300 and the shopkeeper is selling it at a discount of 20%. find out the discount amount and the selling price.

SOLUTION : Market Price = 300, Discount percentage = 20%

For a marked price of Rs 100 the discount is Rs. 20

Therefore, for a marked price of Rs. 1 the discount will be Rs. $\frac{20}{100}$

Discount on Rs. 300 = $\frac{20}{100} \times 300$ = Rs. 60

Selling price = 300 – 60 = 240

EXAMPLE-2. A book was sold for Rs. 45 after a discount of 10%. What was the marked price of the book?

SOLUTION : On a marked price of Rs. 100 the discount is Rs. 10.

Its selling price 100 – 10 = Rs. 90

Comparing-	**Selling price**	**Purchase Price**
	90	100
	45	?

When a book is sold for Rs.90 then it's marked price is Rs 100

If the book is sold for Rs.45 then the marked price is $\frac{100}{90} \times 45$ = Rs. 50

Therefore the marked price would be Rs. 50

Try This



1. A sari with the marked price of Rs. 800 is sold at a discount of 15%. Find out the selling price of the sari?
2. During sale a shop gives a discount of 15% on the marked price of all the goods being sold. What would a customer have to pay for a pair of jeans and a shirt with marked prices Rs.17 50 and Rs 650 respectively?
3. Mahi bought a pair of shoes from a sale on a discounted price of 20%. If he paid Rs.1200 for the shoes, what was the marked price of the shoes?

Sale Tax



Manish and Sahil bought some things from a shop. The shopkeeper gave them the bill. They looked at the bill carefully

BILL

Bill No. : 27 Date : 13.02.15

Name : Manish

S.No.	Name of the Item	Qty.	Rate	Amount
1.	Soap	10	19	190
2.	Toothpaste	1	40	40
3.	Coconut Oil (200 *ml.*)	2	85	170
	Total Amount	-	-	400
	Sales Tax 5%	-	-	20
	Total			420

Sahil : What is this sales tax?

Manish : This is the tax charged by the government on anything sold.

EXAMPLE-3. Rajan bought a cooler for Rs.2700 including sales tax at 8%. Find out the price of the cooler before the tax was added.

SOLUTION : 8% sales tax means that if the price without tax is Rs. 100 then price with tax is Rs. 108.

Now, if the price with taxes is Rs. 108 then the actual price is Rs. 100

Hence when the price with tax is Rs. 2700 then actual price is

$$= \frac{100}{108} \times 2,700$$

= 2,500 Rs.

Try This

1. Whenever you go shopping with your family look at the bill and see which are the products on which the shopkeeper charges sales tax. Make a list of these. Discuss with your friends in your class.





2. If you go to the market to buy something and have a 10% discount but have to pay 5% sales tax on materials purchased, then which of these two will be better for you-

   (i) Take a discount of 10% and add 5% sales tax

   (ii) Add 5% sales tax and then take 10% discount

Exercise - 7.1



1. Mohan saves Rs. 3,950 after spending 75% of his salary. What is his monthly salary?
2. In class 10, 60% are girls. Their number is 18, so what is the ratio of numbers of boys to girls in the class?
3. An item marked at Rs. 950 was sold for Rs. 760. Find the discount and discount percent given?
4. Lalita bought a motorcycle for Rs. 9,016 including 12% sale Tax. What was the cost before adding the sale Tax?
5. A flour mill costs Rs. 1,30,000. The sales tax is taken at 7% on this. If Shweta buys this mill, find the price she paid for it?
6. A machine is sold with 8% discounted price at Rs. 1,748. What is the marked price of the machine?
7. We have two baskets. One has 8 mangoes and 4 watermelons and the other has 14 mangoes and 7 watermelons. find out if the ratio of fruits in both the baskets is the same or different?
8. If 5% sales tax is added to the purchase of the following items, then find the selling price of each:-

   (i) Shirt with price of Rs. 200/-

   (ii) 5 *kg* sugar at the rate of Rs. 30/-*per kg*
9. Sarita bought a cooler for Rs. 5,750/- including a tax of 15%. What was the price of cooler before tax.
10. Suraj brought a television for Rs. 10260/- including a sales tax of 8%. What was the price before tax.

Interest

The extra amount charged for the use of some borrowed amount for a certain period is called interest. It is calculated in two ways:-

1. Simple Interest 2. Compound Interest

Simple Interest

You are familiar with simple interest and how to calculate it. Let us recall through an example:

EXAMPLE-4. Find out the simple interest on Rs. 10,000 for 3 years at 10% annual rate of interest.

SOLUTION. Principal = Rs. 10,000 Rate = 10% Time = 3 years

We know, simple interest $= \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$

$= \frac{10,000 \times 10 \times 3}{100}$

= Rs. 3,000

Interest and Compound Interest



Sudhir borrowed Rs. 2,00,000 for 4 years at 8% annual rate of interest. The condition was that he would have to pay the interest at the end of each year. He was to deposit Rs. 16,000/- as the interest for the first year. He was not able to deposit that for some reason next year he went deposit the earlier due amount and the interest for the second year. Ajay told Sudhir- you have to give Rs. 33,280 as the interest for the two years.

Sudhir said the interest for the two years is (Rs. 16000 + Rs. 16000) Rs. 32,000.

Ajay said- We are not taking any extra amount. The additional amount is the interest on the interest amount that was due but not paid last year. This is called compound interest.

EXAMPLE-5. Sadhna borows Rs. 10,000/- for 3 years at 10% annual rate of interest. She has to return the amount with the extra amount (interest). Find out the compound interest and the total amount to be paid after 3 yeras.

1. Principal amount for the first year, P_1 = Rs. 10,000/-

Interest for the first year $10{,}000 \times \frac{10}{100} = 1{,}000$

SI_1 = Rs. 1,000/-

2. The principal for second year, $P_2 = P_1 + SI_1$

$= 10{,}000 + 1{,}000 =$ Rs. 11,000

Hence interest for second year

$SI_2 = 11{,}000 \times \frac{10}{100} =$ Rs. 1,100/-

3. The principal for third year, $P_3 = P_2 + SI_2$

$= 11{,}000 + 1{,}100$

= Rs. 12,100

The interest for third year $SI_3 = 12{,}100 \times \frac{10}{100}$

= Rs. 1,210

4. Hence amount to be paid $P_3 + SI_3$

$= 12{,}100 + 1{,}210$

= Rs. 13,310

Total interest $= SI_1 + SI_2 + SI_3$

$= 1{,}000 + 1{,}000 + 1{,}210$

= Rs. 3,310

Will the amount of compound interest be different from simple interest?

Lets find out:-

Simple interest for 3 years $= \frac{\text{Principle} \times \text{Time} \times \text{Rate}}{100}$

$= \frac{10{,}000 \times 3 \times 10}{100}$

= Rs. 3,000/-

We see that because of compound interest an extra amount Rs. 310 has to be paid. For simple interest the principle amount remains the same while for compound interest it changes each year.

Let us try to find a formula to calculate the compound interest.

Suppose Principal amount = P Rate = R%

Time = t Total amount = A



The interest for the first year $I_1 = \frac{P_1 \times R \times 1}{100}$

The total amount after first year $\therefore A_1 = P_1 + \frac{P_1 \times R \times 1}{100}$

$$= P_1\left(1 + \frac{R}{100}\right)$$

$= P_2$ (Principal amount for the second year)

Interest for the second year $I_2 = \frac{P_2 \times R \times 1}{100}$ $= P_2\left(\frac{R}{100}\right)$

$$= P_1 \times \left(1 + \frac{R}{100}\right) \times \frac{R}{100}$$

$\therefore$ Total amount after 2 years

$$A_2 = P_2 + I_2$$

$$A_2 = P_1\left(1 + \frac{R}{100}\right) + P_1\left(1 + \frac{R}{100}\right) \times \frac{R}{100}$$

$$= P_1\left(1 + \frac{R}{100}\right) \times \left(1 + \frac{R}{100}\right)$$

$$= P_1\left(1 + \frac{R}{100}\right)^2$$

$= P_3$ (Principle for third year)

Interest for the third year $I_3 = \frac{P_3 \times R \times 1}{100}$ $= P_3\left(\frac{R}{100}\right)$

$$I_3 = P_1\left(1 + \frac{R}{100}\right)^2 \times \frac{R}{100}$$

$\therefore$ Total amount after three years $A_3 = P_3 + I_3$

$$A_3 = P_1\left(1+\frac{R}{100}\right)^2 + P_1\left(1+\frac{R}{100}\right)^2 \times \frac{R}{100}$$

$$A_3 = P_1\left(1+\frac{R}{100}\right)^2 \times \left(1+\frac{R}{100}\right)$$

$$A_3 = P_1\left(1+\frac{R}{100}\right)^3$$



Hence total amount after t years

$$A_n = P_1\left(1+\frac{R}{100}\right)^t$$

In standard form $$A = P\left(1+\frac{R}{100}\right)^t$$

$$\text{Total amount} = \text{Principal amount} \times \left(1+\frac{Rate}{100}\right)^{Time}$$

Compound interest = Total amount – Principal amount

$$C.I. = A - P$$

$$C.I. = P\left(1+\frac{R}{100}\right)^t - P$$

$$C.I. = P\left[\left(1+\frac{R}{100}\right)^t - 1\right]$$

We use this formula to calculate the compound interest.

EXAMPLE-6. Find out the compound interest for 2 years on Rs. 5600/- at 5% rate of annual interest.

SOLUTION : We know A = P $\left(1+\frac{R}{100}\right)^t$

Here, Principal (P) = Rs. 5,600; Rate (R) = 5% annual; Time (t) = 2 years



Hence, $A = 5,600 \times \left(1 + \frac{5}{100}\right)^2$

$A = 5,600 \times \left(1 + \frac{1}{20}\right)^2$

$A = 5,600 \times \left(\frac{21}{20}\right)^2 \quad = 5,600 \times \left(\frac{21}{20}\right) \times \left(\frac{21}{20}\right)$

A = Rs. 6,174/-

$\therefore$ Compound interest = Total amount – Principal amount

= 6,175 – 5,600 = Rs. 574/-

EXAMPLE-7. Shyam deposited Rs. 64,000/- in a nationalised bank. If the annual rate of compounded interest is $2\frac{1}{2}\%$ then what is the total amount he would get after 3 years? What is the interest he would get?

SOLUTION : We know that $A = P\left(1 + \frac{R}{100}\right)^t$

Here, Principal (P) = Rs. 64,000/-

Rate (R) $2\frac{1}{2}\% = \frac{5}{2}\%$ annual; Time (t) = 3 years



Hence, $A = 64,000 \times \left(1 + \frac{5/2}{100}\right)^3$

$A = 64,000 \times \left(1 + \frac{5}{2 \times 100}\right)^3$

$A = 64,000 \times \left(1 + \frac{1}{40}\right)^3$

$A = 64,000 \times \left(\frac{41}{40}\right)^3$

$A = 64,000 \times \left(\frac{41}{40}\right) \times \left(\frac{41}{40}\right) \times \left(\frac{41}{40}\right)$

$A = 41 \times 41 \times 41$ = Rs. 68,921/-

Amount of interest from bank = Total amount – Principal amount

= 68,921 – 64,000 = Rs. 4,921/-

Try This



1. Find out the compounded interest after 2 years on Rs. 15,000/- at an annual rate of interest of 6%.
2. Find out the compounded interest after 3 years on Rs. 8,000/- at an annual rate of interest of $5\frac{1}{2}$%.
3. Find out the total amount and compound interest:-
   (i) On Rs. 10,700/- after 3 years at an annual rate of interest of $14\frac{1}{2}$%.
   (ii) On Rs. 18,000/- after 2 years at an annual rate of interest of 10%.

Calculation when Compounded Quarterly or Half-Yearly

In many situations including in banks the interest is now not calculated annualy but after each quarter or after 6 months. This means the principal charges every 6 months or 3 months.

When interest is compounded half yearly then we have 2 periods of 6 months in each year and the rate of interest is halved. Similarly if the interest is compounded quarterly then each year has 4 quarters and the rate of interest be comes one fourth.

The calculation are done as follows:-

1. For compounding half yearly

$$\text{Total amount} = \text{Principal} \times \left(1 + \frac{\text{half-yearly rate of interest}}{100}\right)^{\text{No. of half years}}$$



For compounding quarterly

$$\text{Total amount} = \text{Principal} \times \left(1 + \frac{\text{quarter rate of interest}}{100}\right)^{\text{No. of quarters}}$$

Try This



1. Calculate the compound interest for one and a half-year Rs. 4,000/- at 5% rate of interest?
2. If interest is compounded half-yearly then find the compound interest for $1\frac{1}{2}$ years on Rs. 1,500/- at 10% annual rate of interest.
3. Deepak borrowed Rs. 80,000/- from a bank for 2 years. If the annual rate of interest is 10% then find the total amount due when:-

   (i) Interest is compounded annually (ii) Interest is compounded half-yearly.

EXAMPLE-8. Find out the compound interest for $1\frac{1}{2}$ years on Rs. 15,625 at annual rate of interest of 8%, if the interest is compounded half yearly.



SOLUTION : The interest is compounded every half-year.

Hence, time (t) = $1\frac{1}{2}$ year = 3 half-year

Rate (R) = 8% annual = $\frac{8}{2}$ % half-yearly = 4% half-yearly

Principal (P) = Rs. 15,625/-

$$\therefore \text{ Total amount} = \text{Principal} \times \left(1+\frac{\text{half-yearly rate of interest}}{100}\right)^{\text{No. of half years}}$$

$$\text{Total amount} = 15,625\times\left(1+\frac{4}{100}\right)^3 = 15,625\times\left(1+\frac{1}{25}\right)^3$$

$$\text{Total amount} = 15,625\times\left(\frac{26}{25}\right)^3 = 15,625\times\left(\frac{26}{25}\right)\times\left(\frac{26}{25}\right)\times\left(\frac{26}{25}\right)$$

$$\text{Total amount} = 26\times 26\times 26 = \text{Rs. } 17,576/-$$

$\therefore$ Compound interest = Total amount – Principal

= 17,576 – 15,625 = Rs. 1,951

EXAMPLE-9. Calculate the interest for a quarterly compounded interest over a peirod of 9 months on Rs. 1,000/- for 8% annual rate of interest.

SOLUTION : Here interest is compounded quarterly

Hence, Time (t) = 9 months = 3 quarters

Rate (R) = 8% annual = $\frac{8}{4}$ % quarterly = 2% quarterly

Principal (P) = Rs. 1,000/-

$$\therefore \text{ Total amount} = \text{ Principal} \times \left(1+\frac{\text{quarterly rate of interest}}{100}\right)^{\text{No. of quarters}}$$

$$\text{Total amount} = 1,000\times\left(1+\frac{2}{100}\right)^3 = 1,000\times\left(1+\frac{1}{50}\right)^3$$

$$\text{Total amount} = 1,000\times\left(\frac{51}{50}\right)^3 = 1,000\times\left(\frac{51}{50}\right)\times\left(\frac{51}{50}\right)\times\left(\frac{51}{50}\right)$$

$$\text{Total amount} = \frac{1,32,651}{125} = \text{Rs. } 1,061.20$$

$\therefore$ Compound interest = Total amount – Principal

= 1,061.20 – 1,000 = Rs. 61.20

EXAMPEL-10. What principal amount would become Rs. 2,809/- after 2 years of compound interest at annual rate of 6%.

SOLUTION : Here, Total amount (A) = Rs. 2,809/- Principal (P) = ?

Rate (R) = 6% annual Time (t) = 2 years

$$\therefore A = P\left(1+\frac{R}{100}\right)^t$$

$$2,809 = P\left(1+\frac{6}{100}\right)^2 = P\left(1+\frac{3}{50}\right)^2$$

$$2,809 = P\left(\frac{53}{50}\right)^2 = P\times\frac{53\times53}{50\times50}$$

$$2,809 = P\times\frac{2,809}{2,500}$$

$$\therefore P = \frac{2,809}{1}\times\frac{2,500}{2,809} = \text{Rs. } 2,500/-$$



EXAMPLE-11. At what rate of interest will the principal amount of Rs. 1,000/- become Rs. 1,331/- after $1\frac{1}{2}$ year of half yearly compounded interest?

SOLUTION : Here, Total amount (A) = Rs. 1,331/- Principal (P) = Rs.1,000

Rate (R) = ? Time (t) = $1\frac{1}{2}$ years = 3 half-yearly

$$\therefore A = P\left(1+\frac{R}{100}\right)^t$$

$$1,331 = 1,000\left(1+\frac{R}{100}\right)^3$$

$$\frac{1,331}{1,000} = \left(1+\frac{R}{100}\right)^3$$

$$\left(\frac{11}{10}\right)^3 = \left(1+\frac{R}{100}\right)^3$$

If the terms on both sides of an equation have the same exponent then the bases are equal.



Hence, $\frac{11}{10} = 1+\frac{R}{100}$

$$\frac{11}{10}-1 = \frac{R}{100}$$

$$\frac{1}{10} = \frac{R}{100}$$

$$R = \frac{100}{10} = 10\% \text{ (half-yearly)}$$

Since rate of interest calcualted is half yearly,

Hence, R = 10 × 2 = 20% (annual)

EXAMPLE-12. Mohan borrowed Rs. 31,250/- at 8% annual interest. After how much time would be have to return Rs. 39,336/-

SOLUTION : Here, Total amount (A) = Rs. 39,366/- Principal (P) = Rs. 31,250

Rate (R) = 8% Time (t) = ?

$$\therefore A = P\left(1+\frac{R}{100}\right)^t$$



$$39{,}366 = 31{,}250\left(1+\frac{8}{100}\right)^t$$

$$\frac{39{,}366}{31{,}250} = \left(1+\frac{2}{25}\right)^t$$

$$\frac{19{,}683}{15{,}625} = \left(\frac{27}{25}\right)^t$$

$$\frac{27\times27\times27}{25\times25\times25} = \left(\frac{27}{25}\right)^t$$

$$\left(\frac{27}{25}\right)^3 = \left(\frac{27}{25}\right)^t$$

The bases on either side of the equation being equal the exponents would also be equal.

Hence t = 3 years.

Exercise - 7.2



1. Find the compound interest and the total amount for the following:-

(i)	Principal = Rs. 7,000	Rate = 10% yearly	Time = 4 years
(ii)	Principal = Rs. 6,250	Rate = 12% yearly	Time = 2 years
(iii)	Principal = Rs. 16,000	Rate = 5% yearly	Time = 3 years

2. Rahul borrowed Rs. 1,25,000/- at 12% annual compound interest from a bank to buy car. What would be the total amount and the amount of interest after 3 years?

3. Find out the compound interest on Rs. 50,000/- deposited in a bank at 15% annual rate after 2 years.

4. Calculate the simple and compound interest at 5% annual rate on Rs. 1,260/- after 2 years?

5. Find the difference between the amount of compound interest and simple interest on a principal amount of Rs. 8,000/- at an annual interest rate of 10% after 2 years.

6. Find out the compound interest after $1\frac{1}{2}$ years on an amount of 3,000/- at 8% annual rate of interest when the compounding of interest is half yearly.

7. How much interest would we get from the bank on Rs. 9,000/- in a year with quarterly compounded interest at 8% annual rate of interest.

8. Find the compound interest on Rs. 3,500/- after one year if compounded half yearly .

9. A man borrowed Rs. 25,000/- what total amount would he pay after one year, if the rate of interest is 20% annual and is compounded half yearly.

10. Find the compounded interest for six months on Rs.10,000 at an annual rate of interest of 12% if the interest is compounded every 3 months.

11. How much money should Surjeet deposit in the Punjab National bank so that he gets Rs.6 615 after two years with the rate of interest 5% annually?

12. What amount would become Rs.18,522 at an annual rate of interest of 10% after one and a half years if the interest is calculated every six months?

13. Kabir borrowed Rs. 15,625 from the Indian bank. He returned Rs.17,576 after 3 years. What was the rate of interest charged by the bank?

14. At what annual rate of interest Rs 6,000 would become Rs.6 ,615 in two years?

15. In how much time would the compounded amount become Rs.9,261 from a principal of Rs.8,000 if the interest is compounded annually at a rate of 5% interest?

16. Ahmed got an interest amount of Rs.5853 on his deposit of Rs.46,875 at an annual rate of interest of 8%. if the interest was compounded every six months then find the time of deposit?

17. What is the principal amount on which the difference between the compound interest and simple interest is Rs. 40, after two years at the rate of annual interest 5%?

The Use of the Formula for Compound Interest

The formula for finding the compound interest is used in many situations.

(i) For finding the increase and decrease of population

(ii) For finding the increase and decrease in price of an item

(iii) For finding the total amount when the rate of interest is different in different years.

(a) If the population of a town increases, then

$$\text{expected population} = \text{current population} \left(1+\frac{\text{Rate}}{100}\right)^{time}$$

(b) If the population of a town decreases, then



$$\text{expected population} = \text{current population}\left(1-\frac{\text{Rate}}{100}\right)^{time}$$

(c) If the price of an object increases, then

$$\text{expected price} = \text{current price}\left(1+\frac{\text{Rate of Increase}}{100}\right)^{time}$$

(d) If the price of an object decreases, then

$$\text{expected price} = \text{current price}\left(1-\frac{\text{Rate of Decrease}}{100}\right)^{time}$$

(e) If the rate of interest in successive years is $R_1\%, R_2\%, R_3\%$.... then,

$$A = P\left(1+\frac{R_1}{100}\right)\left(1+\frac{R_2}{100}\right)\left(1+\frac{R_3}{100}\right).....$$

EXAMPLE-13. The present population of the town is 128000, if the population increases at 5% annually than what would be the population after three years?

SOLUTION : The present population is 128000 Rate of increase = 5% annually

Time = 3 Years Expected population = ?

$\because$ Expected population = Current population $\left(1+\frac{\text{Rate of Increase}}{100}\right)^{time}$

$$\text{Expected population} = 1{,}28{,}000\left(1+\frac{5}{100}\right)^3 = 1{,}28{,}000\left(1+\frac{1}{20}\right)^3$$

$$\text{Expected population} = 1{,}28{,}000\left(\frac{21}{20}\right)^3 = 1{,}28{,}000\times\frac{21}{20}\times\frac{21}{20}\times\frac{21}{20}$$

Expected population = $16 \times 21 \times 21 \times 21 = 1{,}48{,}176$

EXAMPLE-14. Kishore bought a bike at Rs. 55,000. If the bike depreciates at 8% annually than what would be the value of the bike after two years?

SOLUTION : Here the price of the bike = Rs.55,000; Rate of depreciation = 8% annually

Time = 2 Years; Expected Price = ?

$\because$ Expected price = Current price $\times \left(1-\frac{\text{Rate of Depreciation}}{100}\right)^{time}$

$$= 55,000\left(1-\frac{8}{100}\right)^2 = 55,000\left(1-\frac{2}{25}\right)^2$$

$$= 55,000\times\left(\frac{23}{25}\right)^2 = 55,000\times\frac{23}{25}\times\frac{23}{25}$$

$$= 88\times 23\times 23 \text{ = Rs. } 46,552/-$$

EXAMPLE-15. A car is priced at Rs.450,000. The car depreciates at 4% annually for the first two years and 10% annually for the next two years. What is the price of the car after four years?

SOLUTION : Here the price of the car = 4,50,000/-

Rate of depreciation = 4% annually

Time (t_1) = 2 years

And then Rate of depreciation = 10% annually

Time (t_2) = 2 years

Desired Value = ?

$\because$ Desired Value = Original price $\times \left(1-\frac{R_1}{100}\right)^{t_1} \times \left(1-\frac{R_2}{100}\right)^{t_2}$

$$= 4,50,000 \times \left(1-\frac{4}{100}\right)^2 \times \left(1-\frac{10}{100}\right)^2$$

$$= 4,50,000\times\left(\frac{24}{25}\right)^2 \times\left(\frac{9}{10}\right)^2$$

$$= 4,50,000\times\frac{24}{25}\times\frac{24}{25}\times\frac{9}{10}\times\frac{9}{10}$$

$$= 7.2\times 24\times 24\times 9\times 9$$

= Rs. 3,35,923



Try This

1. The present population of the village is 8000. If the population is decreasing at the annual rate of 5%. What would be the population after 3 years?
2. The population of the town is 9600. If the rate of increase of the population is 15% annually, then what would be the population of the town after 2 years?



Instalment Scheme (Purchasing in Instalments)

You would have come across many advertisements of this kind-

"Buy the house of your dreams at minimum interest rate with easy instalments" or "Pay only Rs. 30,000 and take the car of your choice to your house, the rest in easy instalments or buy TV or fridge in Rs 1000 and the rest of the payment in easy monthly/ half yearly/ annual instalments"

Under such schemes people can buy costly items without full payment therefore these schemes enable people to buy costly items comfortably.

In instalment schemes, the customer while purchasing the item pays a part amount on the spot. This is called initial down payment. They subsequently sign a contract takes the item in use. They pay the remaining amount in instalments. This is called the instalment amount. The monthly, quarterly, half yearly or annual instalments are fixed with the consent of the customer and the shopkeeper.

Let us understand the instalment scheme through an example-

Finding the rate of interest of an instalment scheme:-

In the instalment scheme only a part of the total cost is paid by the customer at the time of purchase. The rest is paid in instalments and for this the seller takes some extra amount from the customer. This extra amount is the interest.

EXAMPLE-16. The cash down price of a table is Rs.1000/-. Ramesh buys it for a down payment of Rs.400 and monthly instalments of Rs.310 each. Find the rate of interest in the scheme.

SOLUTION : The cash price of table = Rs. 1000/-

The cash down payment = Rs. 400/-

The due amount for instalments= Rs.1000 - Rs.400 = Rs.600

Suppose that the rate of interest for the instalment scheme is r% annually, then the total amount of Rs.600= Principal+ Interest

$$= 600 + \frac{600 \times r \times 2}{100 \times 12}$$

$$= 600 + r \qquad \text{.....(1)}$$

The total amount after two months of the monthly instalment of Rs.310 paid after the first month

$$= 310 + \frac{310 \times r \times 1}{100 \times 12}$$

$$= 310 + \frac{31r}{120} \qquad \text{.....(2)}$$

The total amount of Rs.310 given after two months=Rs 310

Therefore combining the two monthly instalments at the end of two months

$$\text{Total amount} = 310 + \frac{31r}{120} + 310 = 620 + \frac{31r}{120}$$

Using both (1) and (2)

$$600 + r = 620 + \frac{31r}{120}$$

$$r - \frac{31r}{120} = 620 - 600$$

$$\frac{120r - 31r}{120} = 20$$

$$89r = 20 \times 120$$

$$r = \frac{2,400}{89} = 26.97\% \text{ (approximately)}$$



Therefore, the annual rate of interest in the instalment scheme=26.97%

Using another method:

The cash price of table = Rs.1000

Down cash payment = Rs.400

The amount due in instalments = Rs. 1000 – Rs. 400

= Rs.600

Number of equal instalments = 2

The total amount paid in instalments = 2 × 310 = Rs. 620

The total interest paid = 620 – 600 = Rs. 20

The principal amount for the first month = Rs. 1000 – Rs. 400 = Rs. 600

The principal amount for the second month = 600 – 310 = Rs. 290

Total principal amount (for one month) = 600 + 290 = Rs. 890

Suppose the annual rate of interest is r%, then

$$\text{Total interest} = \frac{890 \times r \times 1}{100 \times 12} = 20$$

$$\text{Or } r = \frac{20 \times 100 \times 12}{890}$$

$$r = \frac{2,400}{89} = 26.97\% \text{ (approximately)}$$

Finding the Instalment Amount

Let us now try to understand how the seller calculates the instalment amount. The shopkeeper buys the product at some price. He knows that using the instalment scheme more items can be sold. Therefore, to get an appropriate interest rate he wants to decide the cash down amount, the instalment amount and the number of instalments.

EXAMPLE-17. The price of a television set is Rs.12,000. This is sold at Rs.3,000 down payment and two equal monthly instalments. If the rate of annual interest is 18% then find the amount of each instalment.

SOLUTION :

The cash price of television = Rs. 12,000

Cash down paid = Rs. 3,000

Payment to be made in instalments = Rs. (12000 – 3000) = 9000

Rate of Interest = 18% Annual

Let each instalment be = Rs. x

$$\text{Interest} = \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$$

$$\text{The amount of one month interest on Rs. 9000} = \frac{9000 \times 18 \times \frac{1}{12}}{100} = \text{Rs. } 135$$

The principal amount after one month = 9000 + 135 = Rs. 9135

The amount to be paid after paying the first instalment = Rs. $(9135 - x)$

The total amount of this amount would be the due instalment after one month.

$$\text{Hence the interest on Rs. } (9135 - x) \text{ for one month} = \frac{(9135 - x) \times 18 \times \frac{1}{12}}{100}$$

$$= \frac{3(9135 - x)}{200} \text{ Rs.}$$

The total amount after one month = Principal + Interest

$$= (9135 - x) + \frac{3(9135 - x)}{200}$$

$$= \frac{200(9135 - x) + 3(9135 - x)}{200}$$

$$= \frac{1827000 - 200x + 27405 - 3x}{200}$$

Hence $$= \frac{1827000 - 200x + 27405 - 3x}{200} = x$$

$$1854405 - 203x = 200x$$

Or $$184405 = 200x + 203x$$

$$403x = 1854405$$

Hence, $$x = \frac{1854405}{403} = \text{Rs. } 4601.50$$



Hence, the amount of each instalment is Rs. 4,601.50

Finding the Cash Price

If for an instalment scheme the amount of each equal instalment, number of instalments, rate of interest and cash down payment amount is given then we can find the cash price of the item. Let us understand this through an example.

EXAMPLE-18. A bicycle is available for Rs.500 cash down and two annual equal instalments of Rs.1210 each. If the rate of interest is 10 % annually then find the cash price of the bicycle.

SOLUTION : Suppose the cash price of the bicycle is Rs. x.

Cash down payhment = Rs. 500

First Instalments = Rs. 1210, Rate of Interest = 10%

The amount due after cash down payment = Rs. $(x - 500)$

The interest for one year on $(x - 500)$ = $\frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$

$$= \frac{(x-500)\times 10 \times 1}{100} = \frac{(x-500)}{10} \text{ Rs.}$$

The total amount after one year = Principal + Interest

$$= (x - 500) + \frac{x - 500}{10}$$

$$= (x - 500)\left(1 + \frac{1}{10}\right) = (x - 500)\frac{11}{10}$$

The amount due after the first instalment (the principal amount for the second year)

$$= [(x - 500)\ \frac{11}{10} - 1210]$$

The total amount at the end of second year = Principal $\left\{1+\frac{\text{Rate}}{100}\right\}^{\text{Time}}$

$$= [(x - 500)\ \frac{11}{10} - 1210]\ [1 + \frac{10}{100}]^1$$

$$= [(x - 500)\ \frac{11}{10} - 1210]\ \frac{11}{10}$$

The total amount at the end of second year is equal to the due instalment for the second year.

$$[(x - 500)\ \frac{11}{10} - 1210]\ \frac{11}{10} = 1210$$

$$[(x - 500)\ \frac{11}{10} - 1210]\ 11 = 12100$$

Or $\quad [(x - 500)\ \frac{11}{10} - 1210] = 1100$

Or $\quad (x - 500)\ \frac{11}{10} = 1100 + 1210$

Or $\quad (x - 500) = 2310 \times \frac{10}{11}$

Or $\quad x - 500 = 2100$

$\quad x = \text{Rs. } 2600$



Hence, the cash price of the bicycle is Rs. 2,600/-

Problems Involving Compound Interest

In a monthly instalment scheme when the total time is less than one year simple interest is used. However sometimes the seller charges compound interest even for less than one year. In such cases interest is compounded half yearly or quarterly.

Sometimes instalments are given for more than one year, in such cases also compound interest is used.

EXAMPLE-19. A fridge with a cash price of Rs.15,000, is available for Rs 2,250 cash down and two equal half yearly instalments at 8% rate of annual interest. If interest is compounded every six months then find the amount of each instalment.

SOLUTION : The cash price of the Fridge = Rs. 15,000

Cash down payment = Rs. 2,250

Remaining payable amount = Rs. (15,000 – 2,250)

= Rs. 12,750

Rate of Interest = 8% annually = 4% half yearly

Let the amount of each half yearly instalment be Rs. x, and P_1 and P_2 the principal amounts for the first and second term.

$$\therefore \quad x = P_1\left(1+\frac{4}{100}\right)^1 \text{ and } x = P_2\left(1+\frac{4}{100}\right)^2$$

$$x = P_1\left(\frac{26}{25}\right) \text{ and } x = P_2\left(\frac{26}{25}\right)^2$$

$$P_1 = \left(\frac{25}{26}\right)x \text{ and } P_2 = \left(\frac{25}{26}\right)^2 x$$



Hence, $$12,750 = \left(\frac{25}{26}\right)x + \left(\frac{25}{26}\right)^2 x$$

$$12,750 = \left(\frac{25}{26}\right)x\left[1+\frac{25}{26}\right]$$

$$12,750 = \left(\frac{25}{26}\right)x\left(\frac{51}{26}\right)$$

$$x = \frac{12,750\times 26\times 26}{25\times 51} = \text{Rs. } 6760/-$$

The amount of each instalment = Rs. 6,760/-

Exercise - 7.3



1. A chair is sold for cash price of Rs. 450 or a cash down payment of Rs.210 and two equal monthly instalments of Rs.125 each. Find out the rate of interest charged in the scheme.
2. A TV stand is sold for Rs.3,000 cash or cash down payment of Rs.600 with two equal monthly instalments of Rs.1,250. Find out the rate of interest charged in the scheme.
3. The cash price of a fan is Rs.1,940. In the instalment scheme it is available for a cash down payment of Rs 620 and two equal monthly instalments. If the rate of annual rate of interest charged in under the scheme is 16% then find the amount of each instalment.

4. The cash price of a microwave oven is Rs. 20,100. if this is available for Rs.3700 cash down payment with two equal monthly instalments at 10% annual rate of interest then find the amount of each instalment.

5. An Iron is bought for cash down payment of Rs. 210 and two equal monthly instalments of Rs.220 each. If the rate of interest is 20% annually then find the cash price of the Iron.

6. The cash price of a scooter in a showroom is Rs.20,000. Under an instalment scheme this is available for cash down payment of Rs.11,000 with two equal annual instalments at 25% annual rate of interest. If the interest is compounded every year then find the amount of each instalment.

7. Washing machine is available for Rs 12,000 cash payment or after a cash down payment of Rs.3600, two equal half yearly instalments. If the annual rate of interest is 20% and the interest is compounded every six months, then find the amount of instalment.

8. A sewing machine is available for Rs 3,000 cash or a cash down payment of Rs 450 and two equal half yearly instalments. If the compound interest is 4% annually then what would be the amount of each instalment.

What Have We Learnt



1. We compare quantities on the basis of ratios and percentages etc.
2. Percentages are used to calculate the amount of discount, sales tax and interest.
3. The reduction in the marked price is called discount.
4. The government charges sales tax on any item sold. The seller adds the sales tax amount to the bill amount.
5. The interest charged is of two types, simple interest and compound interest.
6. (i) If the interest is annually compounded then $A = P\left(1+\frac{R}{100}\right)^t$.

   (ii) When the interest is compounded half yearly then time periods are number of half years and the rate of interest is halved then total amount $A = P\left(1+\frac{R}{200}\right)^{2t}$.

   (iii) When interest is compounded quarterly, the time period are number of quarters and rate is the fourth the annual rate. So $A = P\left(1+\frac{R}{400}\right)^{4t}$.